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Khulna University, Khulna
Mathematics Discipline

4^{th} Year, 2^{nd} Term Examination, Session: 2018-2019
Course No: **Math-4201**, Full Title of Course: **Topology-II**
Full Marks: 60, Time: 03 Hours

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.
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Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions — **Marks**

1. (a) State and prove the generalized Heine-Borel theorem. — 05
   (b) Show that the product space $X_1 \times X_2$ is connected iff both X_1 and X_2 are connected. — 05

2. (a) Prove that a subset A of a topological space is closed iff no net in A converges to a point in $(X\text{-}A)$. — 05
   (b) Prove that a topological space $(X, \mathfrak{I})$ is compact iff each set in X has a cluster point. — 05

3. (a) Let $\mathcal{F}$ be a filter on a non-empty set X and let $A \subset X$. Show that there exists a filter F_1 finer than $\mathcal{F}$ such that $A \in F_1$ iff $A \cap F \neq \phi,\ \forall F \in \mathcal{F}$. — 05
   (b) Prove that a compact Hausdorff space is separable and metrizable if it is second countable. — 05

4. (a) Show that the projection map of a product space are continuous and open. — 05
   (b) State and prove the Urysohn metrization theore. — 05

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions — **Marks**

5. (a) State and prove the Baire Category theorem. — 05
   (b) Show that local compactness is a topological property. — 05

6. (a) State and prove the Cantor's intersection theorem. — 05
   (b) Show that the set of complex numbers with usual metric is complete metric space. — 05

7. (a) Let A be a closed subset of $C[0,1]$. Then show that the following are equivalent: — 05
      (i) A is compact
      (ii) A is uniformly bounded and equicontinuous
   (b) Let $< f_n >$ be a sequence of continuous functions from a topological space X into a metric space Y and let $< f_n >$ converge uniformly to $g: X \to Y$. Show that g is continuous. — 05

8. (a) If $f: I \to R$ be continuous on $I = [0,1]$, then show that f is uniformly continuous. — 05
   (b) Show that a compact T_2 space has only one uniformly compatible with its topology. — 05

****** THE END ******

Khulna University, Khulna

Mathematics Discipline

4th Year, Term-II Examination, Session: 2020-2021

Course No: **Math-4201**

Full Title of Course: **Topology**

Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Define topological space. Also describe discrete, indiscrete topology and usual topology. 03

   (b) Let $X=\{a,b,c,d,e\}$ and $T_1=\{X,\phi,\{a\},\{a,b\},\{a,c\}\}$, $T_2=\{X,\phi,\{a,b,c\},\{a,b,d\},\{a,b,c,d\}\}, T_3=\{X,\phi,\{a\},\{a,b\},\{a,c,d\},\{a,b,c,d\}\}$. Determine whether or not T_1, T_2 and T_3 forms a topological space on X. 03

   (c) Let T_1 and T_2 be any two topological spaces on X. Prove that $T_1 \cap T_2$ is also a topology on X. Also given two topologies T_1 and T_2 on $X=\{a,b,c\}$ so that $T_1 \cup T_2$ is not a topology on X. 04

2. (a) A subset of A of a topological space (X,T) is open iff $A^0 = A$. 03

   (b) Determine whether or not the class of open equilateral triangles and the class of open squares forms a base of the plane R^2. 03

   (c) Let $A=\{b,c,d\}$ be a subset of $X=\{a,b,c,d,e\}$. Consider the topology $T_1=\{X,\phi,\{a\},\{c,d\},\{a,c,d\},\{b,c,d,e\}\}$, find the interior, exterior and boundary of A. 04

3. (a) Define continuous function between two topological spaces X and Y. Prove that $|x|$ is continuous. 03

   (b) Draw the graphs of the mapping $\Pi: R^2 \to R$ by $\Pi_1(\langle x,y\rangle)=x$ *and* $\Pi_2(\langle x,y\rangle)=y$. Prove that Π_1 and Π_2 are continuous. 03

   (c) Prove that $f: X \to R$ Defined by $f(x)=\tan\frac{1}{2}\pi x$ is homeomorphic. Also show that length, boundedness and Cauchy sequence are not topological invariant. 04

4. (a) Find the difference between normed spaces and metric spaces. 02

   (b) A topological spaces (X,T) is a T_1-space. 04

   (c) Prove that each singleton set in a Hausdorff space is closed. 04

Section-B

There are **FOUR** questions in these sections. Answer any **THREE** questions

5. (a) Define Hausdroff space. Find the difference between Hausdorff space and T_1-space. Prove that every metric space is a Hausdorff space. 04

   (b) Consider the topology $T_1=\{X,\phi,\{a\},\{b\},\{a,b\}\}$ on the set $X=\{a,b,c\}$. Prove that (X,T) is a normal space but not T_1-space. 03

   (c) Prove that a continuous image of a sequentially compact set is sequentially compact. 03

6. (a) Let (X,d) be a complete metric space and $Y \subset X$. Show that Y is complete iff Y is closed. 03

   (b) State Hein-Borel theorem. Prove that (0,1) interval on R is not compact. 03

   (c) Show that a closed subset of a compact space is compact. 04

7. (a) Define product space. Prove that the usual topology on $R^2 = R \times R$ is the product topology. 02

   (b) Let, $X=\Pi_i X_i$ where $\{X_i : i \in I\}$ be a collection of Hausdorff space. Prove that X is Hausdorff space. 04

   (c) Prove that a function f form a topological space Y into a product space $X=\Pi_i X_i$ is continuous if and only if for every projection Π_i , the composition mapping Π_i of : $Y \to X_i$ is also continuous. 04

8. (a) Define separated sets and give two examples. 03

   (b) Prove that the product space $X_1 \times X_2$ is connected iff both X_1 and X_2 are connected. 04

   (c) Let $A=\{(x,y): x^2-y^2 \geq 6\}$ and let G and H be defined by $G=\{(x,y): x<-2\}$, and $H=\{(x,y): x>2\}$ respectively. Prove that A is disconnected. 03

Khulna University, Khulna
Mathematics Discipline
4^{th} Year, Term-II Examination, Session: 2020-2021
Course No: **Math-4201**
Full Title of Course: **Topology**
Full Marks: 60, Time: 03 Hours

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- Use separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions — Marks

1. (a) Define topological space. Also describe discrete, indiscrete topology and usual topology. 03

(b) Let $X=\{a,b,c,d,e\}$ and $T_1=\{X,\phi,\{a\},\{a,b\},\{a,c\}\}$, $T_2=\{X,\phi,\{a,b,c\},\{a,b,d\},\{a,b,c,d\}\}, T_3=\{X,\phi,\{a\},\{a,b\},\{a,c,d\},\{a,b,c,d\}\}$. Determine whether or not T_1, T_2 and T_3 forms a topological space on X. 03

(c) Let T_1 and T_2 be any two topological spaces on X. Prove that $T_1 \cap T_2$ is also a topology on X. Also given two topologies T_1 and T_2 on $X=\{a,b,c\}$ so that $T_1 \cup T_2$ is not a topology on X. 04

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(c) Prove that each singleton set in a Hausdorff space is closed. 04

Section-B

There are **FOUR** questions in these sections. Answer any **THREE** questions

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(b) Consider the topology $T_1=\{X,\phi,\{a\},\{b\},\{a,b\}\}$ on the set $X=\{a,b,c\}$. Prove that (X,T) is a normal space but not T_1-space. 03

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Khulna University, Khulna
Mathematics Discipline
4th Year, 2nd Term, Examination; Session: **2021-2022**
Course No: **Math-4201,** Full Title of Course: **Topology**
Full Marks: **60**, Time: **03 Hours**

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- Use separate sheet for each section.
- All the symbols have their usual meanings.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

1. (a) Briefly discuss about topological space, accumulation points, closed sets and neighborhoods of a point with relevant example. 02

(b) Let $f: X \to Y$ be a function from a non-empty set X into a topological space (Y, u). Again, let τ be the class of inverse of open subsets of $Y: \tau = \{f^{-1}[G]: G \in u\}$. Show that τ is a topology on X. 04

(c) Consider the topology $\tau = \{\Phi, \{x_1\}, \{x_1, x_2\}, \{x_1, x_3, x_4\}, \{x_1, x_2, x_3, x_4\}, \{x_1, x_2, x_5\}, X\}$ on $X = \{x_1, x_2, x_3, x_4, x_5\}$. Determine the derived sets of $i) A = \{x_3, x_4, x_5\}$ and $ii) B = \{x_2\}$. 04

2. (a) Explain base, subbase, local base, first axiom space, completely separable space and hereditary properties. 03

(b) To show that a space whose topology has a countable base is separable. 04

(c) Let (X, τ) be topological space where $X = \{x, y, z, u\}$ and $\tau = \{\Phi, \{x\}, \{y\}, \{x, y\}, X\}$. Show that X is a first countable space. 03

3. (a) Define metric space. Give an example which is not metric space. 03

(b) If d is a metric on X, then prove that $\rho(a,b) = \dfrac{d(a,b)}{1+d(a,b)}$ is also metric on X. 03

(c) If $1 < p < \infty$ and $\dfrac{1}{p} + \dfrac{1}{q} = 1$, then prove that $ab \leq \dfrac{a^p}{p} + \dfrac{b^q}{q}$. 04

4. (a) Define Tychonoff space, Hausdorff space, normal and regular space with example. 02

(b) Prove that a topological space (X, T) is a T_1-space iff $\{x\}$ closed for each $x \in X$. 04

(c) Show that the property of a space being regular is hereditary property. 04

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Define Hausdorff space, regular space and normal spaces. Find the relation in a diagram. 04

(b) Prove that every metric space is Hausdorff space. 03

(c) State Urysohn's lemma. Prove that every T_4-space is a T_3-space. 03

6. (a) Let $B = \{D_p : p \in Z \times Z\}$ where D_p has centre p and radius 1 and $1/2$ respectively. Will B cover R^2? Justify your answer. 03

(b) Prove that continuous image of a compact set is compact. 04

(c) State Heine-Borel theorem. Let A be any subset of a topological space X. Then prove that A is compact. 03

7. (a) Discuss product space with example. 02

(b) If (X_1,τ_1) and (X_2,τ_2) are any two topological spaces and let B_1, B_2 be two bases for τ_1 and τ_2 respectively with $X = X_1 \times X_2$. Then show that $B = \{b_1 \times b_2 : b_1 \in B_1, b_2 \in B_2\}$ is a base for the product topology τ on X. 05

(c) Consider the topology $\tau = \{X, \Phi, \{a\}, \{b,c\}\}$ on $X = \{a,b,c\}$ and the topology $\tau^* = \{Y, \Phi, \{u\}\}$ on $Y = \{u,v\}$. Find the defining subbase of the product topology on $X \times Y$. 03

8. (a) Define connected set and separated sets with figure. 03

(b) Define path and homotopic path with figure. 03

(c) Prove that every arcwise connected sets are connected but the converse is not true. 04

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Khulna University, Khulna
Mathematics Discipline
4th Year, 2nd Term, Examination; Session: **2021-2022**
Course No: **Math-4201,** Full Title of Course: **Topology**
Full Marks: **60**, Time: **03 Hours**

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   (c) Let (X, τ) be topological space where $X = \{x, y, z, u\}$ and $\tau = \{\Phi, \{x\}, \{y\}, \{x, y\}, X\}$. Show that X is a first countable space. **03**

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Khulna University, Khulna
Mathematics Discipline
4^{th} **Year**, 2^{nd} **Term**, Examination; Session: **2021-2022**
Course No: **Math-4201**, Full Title of Course: **Topology**
Full Marks: **60**, Time: **03 Hours**

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   (c) Consider the topology $\tau = \{\Phi, \{x_1\}, \{x_1, x_2\}, \{x_1, x_3, x_4\}, \{x_1, x_2, x_3, x_4\}, \{x_1, x_2, x_5\}, X\}$ on $X = \{x_1, x_2, x_3, x_4, x_5\}$. Determine the derived sets of $i)\ A = \{x_3, x_4, x_5\}$ and $ii)\ B = \{x_2\}$. — 04

2. (a) Explain base, subbase, local base, first axiom space, completely separable space and hereditary properties. — 03

   (b) To show that a space whose topology has a countable base is separable. — 04

   (c) Let (X, τ) be topological space where $X = \{x, y, z, u\}$ and $\tau = \{\Phi, \{x\}, \{y\}, \{x, y\}, X\}$. Show that X is a first countable space. — 03

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	(b)	Prove that continuous image of a compact set is compact.	04
	(c)	State Heine-Borel theorem. Let A be any subset of a topological space X. Then prove that A is compact	03
7.	(a)	Discuss product space with example.	02
	(b)	If (X_1, τ_1) and (X_2, τ_2) are any two topological spaces and let B_1, B_2 be two bases for τ_1 and τ_2 respectively with $X = X_1 \times X_2$. Then show that $B = \{b_1 \times b_2 : b_1 \in B_1, b_2 \in B_2\}$ is a base for the product topology τ on X.	05
	(c)	Consider the topology $\tau = \{X, \Phi, \{a\}, \{b,c\}\}$ on $X = \{a,b,c\}$ and the topology $\tau^* = \{Y, \Phi, \{u\}\}$ on $Y = \{u, v\}$. Find the defining subbase of the product topology on $X \times Y$.	03
8.	(a)	Define connected set and separated sets with figure.	03
	(b)	Define path and homotopic path with figure.	03
	(c)	Prove that every arcwise connected sets are connected but the converse is not true.	04

****** THE END ******

Khulna University

Science, Engineering and Technology School
Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
4th Year, Term-II Examination-2024, Session: **2022-2023**
Course Code: **Math-4201**
Course Title: **Topology**
Total Marks: 60, Time: 03 Hours, Credit: 03

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- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

			Marks
1.	**(a)**	Define topological space. Also give example of usual topology, discrete and indiscrete topology.	03
	(b)	Let $X=\{a,b,c,d,e\}$, write down three subclasses of X which will form topology. Check whether it is a topology or not? Find the derived set of $A=\{a,b,c\}$ w.r.to the topology T.	03
	(c)	Define interior, exterior and boundary point. Let $T=\{X,\phi,\{a\},\{c,d\},\{a,c,d\},\{b,c,d,e\}\}$ be a topology on $X=\{a,b,c,d,e\}$. Find the interior, exterior and boundary point of $A=\{a,b,c\}$.	04
2.	**(a)**	Show that the closure of a set is union of that set and its derived set.	05
	(b)	In any topological space, show that $(A\cap B)^0 = A^0\cap B^0$.	05
3.	**(a)**	Define continuous function with respect to some topology with example.	03
	(b)	Draw the diagram and prove that the function $f(x)=\lvert x\rvert$ is continuous.	03
	(c)	Define homeomorphic spaces and give examples. Also define topological property and give two examples which are not topological property.	04
4.	**(a)**	Define metric space, usual metric and trivial metric. Give an example which is not metric space.	03
	(b)	Let $C[0,1]$ denote the collection of continuous functions on $[0,1]$ and let $d(f,g)=\int_0^1\lvert f(x)-g(x)\rvert dx$ and $d^*(f,g)=\sup\{\lvert f(x)-g(x)\rvert : x\in[0,1]\}$. Draw their graphs and are they metric?	03
	(c)	Define open sphere. Graphically explain three unusual spheres.	04

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

			Marks
5.	**(a)**	Define T_1,T_2,T_3,T_4 spaces with examples. Find the relation among them in a diagram.	04
	(b)	Prove that every metric space is Hausdorff space.	03
	(c)	State Urysohn's lemma. Prove that every T_4-space is also a T_3-space.	03
6.	**(a)**	Show that every subspace of a regular space is regular.	05
	(b)	Prove that a continuous image of a compact space is compact.	05
7.	**(a)**	Define product space with example. Consider the topology $T=\{X,\phi,\{a\},\{b,c\}\}$ on $X=\{a,b,c\}$ and the topology $T^*=\{Y,\phi,\{u\}\}$ on $Y=\{u,v\}$. Determine the subbase and base of the product topology $X\times Y$.	03
	(b)	Let $\{X_i : i\in I\}$ be a collection of Hausdorff spaces and let $X=\Pi_i X_i$ prove that X is a Hausdorff space.	03
	(c)	Prove that a function f from a topological space Y into a product space $X=\Pi_i X_i$ is continuous iff for every projection π_i such that $\pi_i\circ f: Y\to X_i$ is also continuous.	04
8.	**(a)**	Define connected set and separated sets with figure.	03
	(b)	Define path and homotopic path with figure.	03
	(c)	Explain the Cantor set with its properties.	04

****** THE END ******

Khulna University

Science, Engineering and Technology School

Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**

Year: **Fourth**, Term: **II** Examination-2025; Session: **2023-2024**

Course Code: **Math-4201**, Course Title: **Topology**

Total Marks: 60, Time: **03** Hours, Credit: 03

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Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

1. **(a)** Define topological space. Also give example of usual topology, discrete and indiscrete topology. **03**

   (b) Let $X = \{a,b,c\}$. List of all topologies of X. From this prove the intersection of two topologies is again a topology. **03**

   (c) Define interior, exterior and boundary point. Let $T = \{X, \phi, \{a\}, \{c,d\}, \{a,c,d\}, \{b,c,d,e\}\}$ be a topology on $X = \{a,b,c,d,e\}$. Find the interior, exterior and boundary point of $A = \{c,d,e\}$. **04**

2. **(a)** Define coarser and finer topology. Consider the topology $T = \{X, \phi, \{a\}, \{c,d\}, \{a,c,d\}, \{b,c,d,e\}\}$ on $X = \{a,b,c,d,e\}$. Find the relative topology of the subset $A = \{a,d,e\}$. **04**

   (b) Define base for a topology. Let (X, D) be the discrete topology on X. Find its base. **03**

   (c) Suppose there are classes of open equilateral triangles and open squares. Draw and determine whether the classes of subsets forms a base for the usual topology on R^2. **03**

3. **(a)** If (X,d) be metric space and $A \subset X$, then prove that $\bar{A} = \{x \in X : d(x, A) = 0\}$. **04**

   (b) Let B be a class of subsets of a non-empty set X. Then show that B is a base for some topology on X iff it satisfies the followings: **06**
   $X = \cup \{b : b \in B\}$ (ii) For any $b, b^* \in B$, $b \cap b^*$ is the union of members of B.

4. **(a)** Show that, a subset A of a metric subspace (Y,d) of a metric space (X,d) is open in Y $iff \, \exists$ an open set G in (X,d) s.t. $A = Y \cap G$. **05**

   (b) In any metric space, prove that any closed sphere is a closed set. **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5.	**(a)**	$(X, \Im)$ is a Hausdroff space *iff* every convergent sequence in X has a unique limit, prove this where $(X, \Im)$ is a first countable space.	**05**
	(b)	Show that a compact Hausdorff space is normal.	**05**
6.	**(a)**	Define open cover and give an example which is not a open cover. State Hein-Borel theorem. Prove that every finite subset of a topological space is compact.	**03**
	(b)	Define finite intersection property. When a subset of a compact space will be compact? Justify it by a theorem with proof.	**04**
	(c)	Prove that $(0,1)$ interval is not compact.	**03**
7.	**(a)**	Define separated and connected sets. Let $A = \{(x,y): x^2 - y^2 \geq 4\}$ and two open sets $G = \{(x,y): x < -1\}$ and $H = \{(x,y): x > 1\}$. Draw their graphs and show that A is disconnected.	**03**
	(b)	Graphically explain the Cantor set and its properties.	**03**
	(c)	Define path and homotopic path with figure.	**04**
8.		Show that: i) a subset of a real line is connected *iff* it is an interval. ii) a closed subspace of a normal space is a normal space.	**10**

****** THE END ******

Khulna University

Science, Engineering and Technology School

Discipline: **Mathematics,** Program: **B.Sc. (Hons.)**

Year: **Fourth,** Term: **I** Examination-2026; Session: **2024-2025**

Course Code: **0541 12 Math 4105**

Course Title: **Topology**

Total Marks: **60,** Time: **03** Hours, Credit: **03**

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

1. **(a)** Define topological space. Also give example of usual topology, discrete and indiscrete topology. [L1; CLO1] **03**

(b) Let $X = \{a,b,c,d,e\}$, write three subclasses of X which will form topology. Check whether it is a topology or not? Find the derived set of $A = \{a,b,c\}$ with respect to the topology T. [L1; CLO2] **03**

(c) Define interior, exterior and boundary point. Let $T = \{X, \phi, \{a\}, \{c,d\}, \{a,c,d\}, \{b,c,d,e\}\}$ be a topology on $X = \{a,b,c,d,e\}$. Find the interior, exterior and boundary point of $A = \{a,b,c\}$. [L2; CLO3] **04**

2. **(a)** Define limit point. Find the limit point of set of rational numbers. [L2; CLO3] **02**

(b) State and prove Minkowski's inequality. [L3; CLO4] **04**

(c) Define normed space. Let $I = [0,1]$, $\|f\| = \sup\{|f(x)|\}$. Prove that it is a norm on $C[0,1]$. [L3; CLO4] **04**

3. **(a)** Define continuous function with respect to some topology with example. [L2; CLO3] **03**

(b) Draw the diagram and prove that the function $f(x) = |x|$ is continuous. [L5; CLO4] **04**

(c) Define homeomorphic spaces and give examples. Also define topological property and give two examples which are not topological property. [L3; CLO4] **03**

4. **(a)** Define metric space, usual metric and trivial metric. Give an example which is not metric space. [L3; CLO4] **03**

(b) Let $C[0,1]$ denote the collection of continuous functions on $[0,1]$ and let $d(f,g) = \int_0^1 |f(x) - g(x)|\,dx$ and $d^*(f,g) = \sup\{|f(x) - g(x)| : x \in [0,1]\}$. Draw their graphs. Are they metric? [L1; CLO4] **04**

(c) Define open sphere. Graphically explain three unusual spheres. [L3; CLO3] **03**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

5.	**(a)**	Define product space. Consider the topology $T = \{X, \phi, \{a\}, \{b,c\}\}$ on $X = \{a,b,c\}$ and the topology $T^* = \{Y, \phi, \{u\}\}$ on $Y = \{u,v\}$. Determine the subbase and base of the product topology $X \times Y$. [L3; CLO5]	**04**
	(b)	Prove that the usual topology on $R^2 = R \times R$ is the product topology. [L2; CLO5]	**03**
	(c)	Topologically draw the graph of unit square, annulus, torus and solid torus. [L4; CLO5]	**03**
6.	**(a)**	Let $B = \{D_p : p \in Z \times Z\}$ where D_p has centre p and radius 1 and $1/2$ respectively. Will B cover R^2? Justify your answer. [L2; CLO6]	**03**
	(b)	Prove that continuous image of a compact set is compact. [L1; CLO6]	**03**
	(c)	State Heine-Borel theorem. Let A be any subset of a topological space X. Then prove that A is compact. [L5; CLO6]	**04**
7.	**(a)**	State and Prove the Cantor's intersection theorem. [L5; CLO4]	**05**
	(b)	Show that a closed subspace of a normal space is a normal space. [L2; CLO4]	**05**
8.	**(a)**	Define connected set and separated sets with figure. [L3; CLO7]	**03**
	(b)	Define path and homotopic path with figure. [L3; CLO8]	**03**
	(c)	Prove that $(0,1)$ interval is not compact. [L2; CLO8]	**04**

****** THE END ******